ETL Transformation Example: Addressing Data Quality Issue

Explore how to transform extracted data to meet business needs by converting raw data into tabular format, eliminating duplicate rows, and resolving data quality issues such as missing values. Understand the practical steps to prepare data for loading into a PostgreSQL database, enhancing the reliability of your ETL pipelines.

Transform

Now that we have extracted the raw data, let's transform it according to the needs and context of the business. At this point, we need to talk to the user who is requesting the data. For this demonstration, the user is the company's data scientist.

The data scientist requests that the data be in *tabular form without missing or null values*. Also, there shouldn't be *any duplicate dates* in the file, and the data needs to have *eight columns separated by commas*. The columns are:

1. "Date"
2. "First_Lottery_number"
3. "Second_Lottery_number"
4. "Third_Lottery_number"
5. "Fourth_Lottery_number"
6. "Fifth_Lottery_number"
7. "Sixth_Lottery_number"

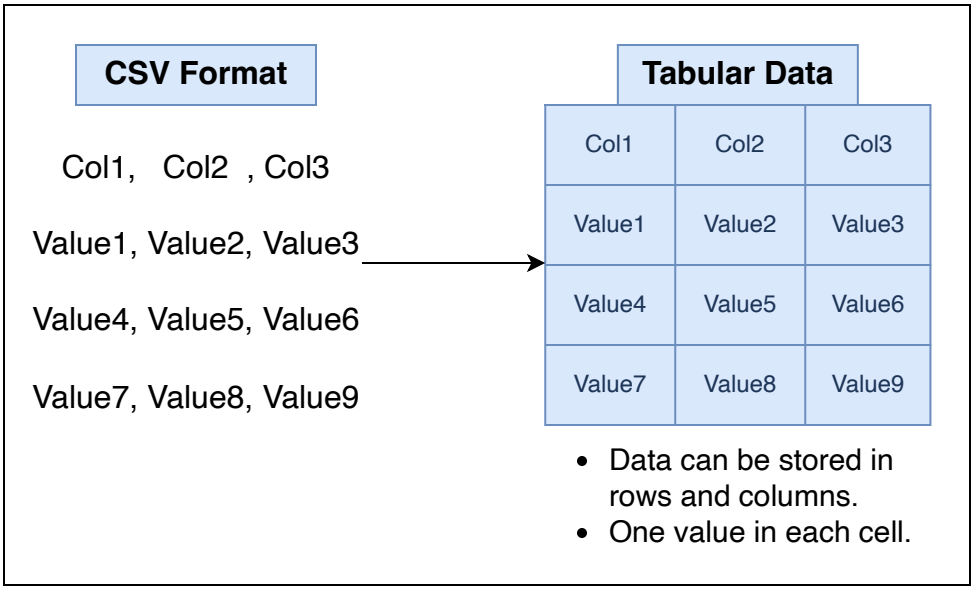
Ask

8. “Multiplier”

Furthermore, they would like us to *sort the “Date” column* in descending order.

Task 1: Make the data tabular

The first request is for the data to be tabular. Tabular data can be organized in a table with rows and columns, and each cell has exactly one value. A popular format for storing tabular data is the CSV format.



The CSV format is used to store tabular data

Since the data's destination is a PostgreSQL relational database, we need to transform it to be tabular. Currently, the second column (“Winning Numbers”) has six values separated by spaces in each cell, violating the tabular data requirements.

Ask

Transforming raw_data.csv to be tabular

To complete the first request, we need to replace all spaces in the second column with commas, essentially making them separate columns. Finally, we add headers to each of these new columns.

transform_data_1... Shell

```shell
#!/bin/bash
# ***                                                    *** #
# *** Issue #1 - Make the data tabular                   *** #
# ***                                                    *** #
echo "Task #1 - Making the Data Tabular"
tabulate_data(){

    # Replace space with a comma
    sed 's/ /,/g' raw_data.csv > temp.csv; mv temp.csv raw_data.csv

    # Delete the headers row
    sed -i 1d raw_data.csv
```

Making the data tabular in the transform_data_1_tabulate.sh file

We've created a file called `transform_data_1_tabulate.sh` to perform these operations. Let's walk through the code line by line.

- **Line 9:** We replace all spaces with commas and append output to a temp file called `temp.csv` using the `sed 's/ /,/g' raw_data.csv > temp.csv` command.
    - We then change the name of `temp.csv` to `raw_data.csv` using the `mv temp.csv raw_data.csv` command.
- **Line 12:** We delete the first row (the header).
- **Line 15:** We create a new file called `temp.csv`.
- **Line 18:** We print a new set of headers into `temp.csv`.
- **Line 19:** We append the content of `raw_data.csv` to `temp.csv`.

Ask

- **Line 22:** We change the name of `temp.csv` to `raw_data.csv`.

Next, we need to update `main.sh` to include the `transform_data_1_tabulate.sh` file.

Click the “Run” button in the code widget below to execute the `main.sh` file.

Files

main.sh

transfor...

extract_...

raw_dat...

main.sh Shell

```
#!/bin/bash

# Setting the project location as an environment variabl
export WORKDIR='/usercode/ETL_Lottery'

# Changing directories to the project folder
cd $WORKDIR

# Extract Raw Data
bash extract_data.sh

# Issue #1 - Make the Data Tabluar
bash transform_data_1_tabulate.sh
```

Run

ETL pipeline: main.sh

Task 2: Eliminate rows with duplicate dates

The next request is that we eliminate rows with duplicate dates. According to the data scientist, there should be no duplicate dates in the file because there are never two lottery results on the same day.

transform_data_2... Shell

Ask

```
#!/bin/bash
# ***                                                    *** #
# *** Issue #2 - Get rid of all rows with duplicate data *** #
# ***                                                    *** #
```

```
echo -e "\nTask #2 - Searching for Duplicate Values"
delete_duplicates(){

    # Grab the duplicate dates into an array
    duplicate_dates=$(cut --delimiter "," --fields 1 raw_data.csv | uniq

    # Check if the array is bigger than 0
    if (( ${#duplicate_dates[@]} > 0 ))
```

Removing duplicate rows: transform_data_2_duplicates.sh

We’ve created a file called `transform_data_2_duplicates.sh` to perform these operations. Let’s walk through the code line by line.

- **Line 10:** We generate an array of duplicate dates from `raw_data.csv`.
- **Lines 13–23:** We check if the size of the array is bigger than `0`. If so, we append all the dates that don't have duplicates to the `temp.csv` file. Then, we change the name of `temp.csv` to `raw_data.csv`.

Next, we need to update `main.sh` to include `transform_data_2_duplicates.sh`.

Click the “Run” button in the code widget below to execute the `main.sh` file.

Files

main.sh
transfor…
transfor…
extract_…
raw_dat…

main.sh Shell

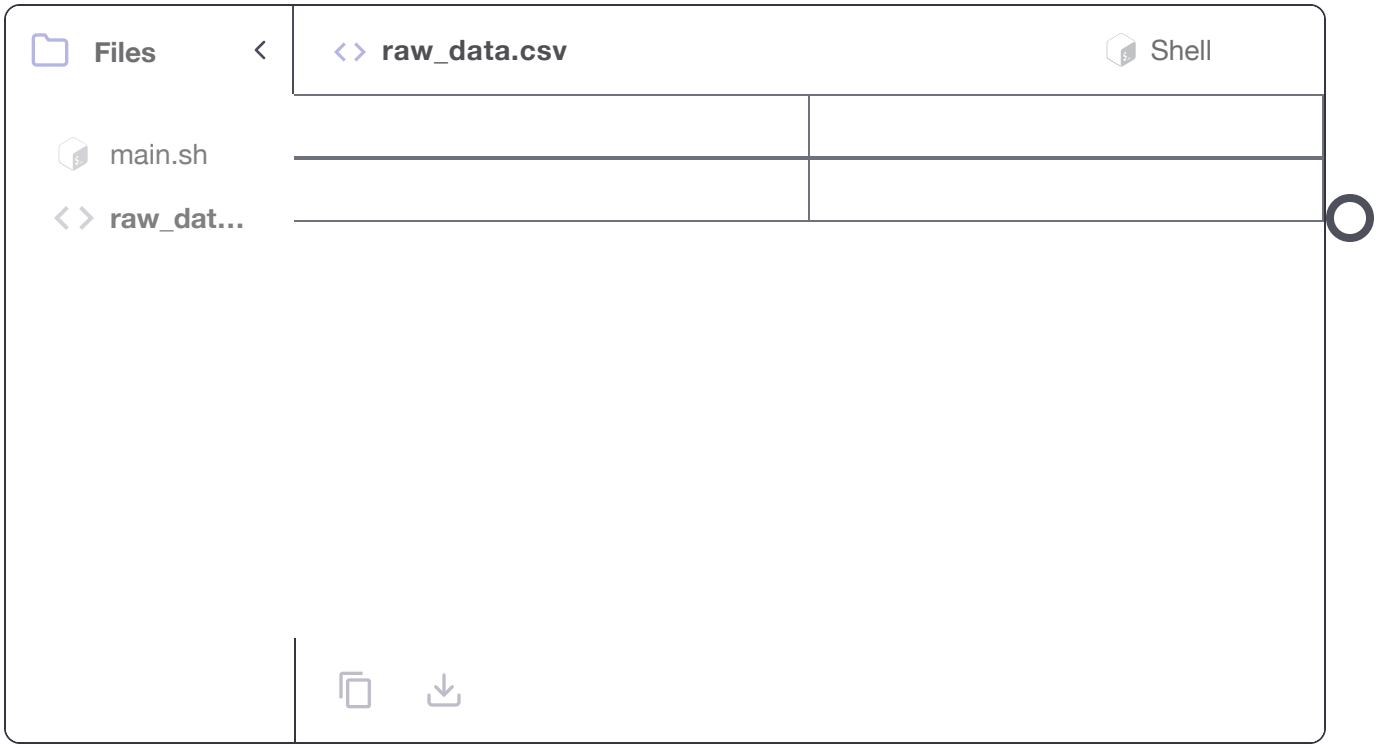
```
#!/bin/bash

# Setting the project location as an environment variabl
export WORKDIR='/usercode/ETL_Lottery'

# Changing directories to the project folder
cd $WORKDIR

# Extract Raw Data
bash extract_data.sh

# Issue #1 - Make the Data Tabluar
bash transform_data_1_tabulate.sh
```

Ask

Run

main.sh - ETL pipeline

Let's check what the data looks like now and what other issues we can spot.



raw_data.csv

It looks like we got rid of rows with duplicate dates. However, there are still a few issues to take care of such as empty rows or rows with `null` values and missing values in the `multiplier` column.

Ask